



BUILD AGENTIC AI SYSTEMS · COHORT 1

Your Weekly Workbook

A day-by-day plan for the 6 days between each live session. Small reps, every day — so you walk into Friday with something to show.

How this works

- › We meet **live every Friday, 5:00–6:30 PM IST**. This workbook covers the days in between.
- › Each week has the **same daily rhythm** — it becomes a habit. Only the build task changes.
- › Every day is **30–45 minutes**. Total $\approx$ 3.5 hrs/week. Don't cram it into Thursday night.
- › Each week ends with one thing to **bring and show** on Friday. That's how we track you.

Your daily rhythm

- › **Sat** — Watch the module, note 3 takeaways
- › **Sun** — Rest / catch-up buffer
- › **Mon** — Guided build in the starter repo
- › **Tue** — Extend it (+ office hours if stuck)
- › **Wed** — Make it yours: your own use-case
- › **Thu** — Test, post a screenshot, write your question
- › **Fri** — Live session: show your work

3 golden rules

- › **Done beats perfect.** Bring something rough every week — we fix it together.
- › **Post your blocker by Thursday** in `#help`. Don't suffer in silence for 6 days.
- › **Build on YOUR use-case.** Every exercise should move your real agent forward.

The one habit that makes this work

Open your laptop for 40 minutes a day, six days a week. That's the whole secret. Students who do the daily reps ship a working agent; students who wait for the weekend don't. Tick each box as you go. 🚀

WEEK 1

Setup & Module 1 — What an agent is

Complete before your **Fri 26 Jun** session · ~3.5 hrs across the week



This week's goal: Your tools work, and you understand the agent loop.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 20 Jun	Watch	Watch Module 1. Note: what are the 4 parts of an agent?	45 min	☐
Sun 21 Jun	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 22 Jun	Build	Fork the starter repo, install it, add your API key to <code>.env</code> .	40 min	☐
Tue 23 Jun	Build +	Run <code>python run.py "hello"</code> . Open <code>agent/loop.py</code> and write a one-line comment on each step.	40 min	☐
Wed 24 Jun	Make it yours	Write your use-case (the real job your agent will do) into <code>config.py</code> and the README.	45 min	☐
Thu 25 Jun	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 26 Jun	LIVE 5 PM	Show your work. Your agent saying hello in the terminal (screenshot) + your one-sentence use-case.		★

 **Bring to Friday:** Your agent saying hello in the terminal (screenshot) + your one-sentence use-case.

Done when:

☐ Repo runs

☐ API key works

☐ Use-case written down

WEEK 2

Module 2 — Your first tool-calling agent

Complete before your **Fri 03 Jul** session · ~3.5 hrs across the week



This week's goal: You can read and run the tool-calling loop end to end.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 27 Jun	Watch	Watch Module 2. Note: how does the model ask for a tool, and how do we answer?	45 min	☐
Sun 28 Jun	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 29 Jun	Build	Run a goal that triggers a tool: <code>python run.py "what is today's date?"</code> . Watch the loop print each step.	40 min	☐
Tue 30 Jun	Build +	Add a print inside the loop to show every tool call + result. Make both example tools fire.	40 min	☐
Wed 01 Jul	Make it yours	List the 2–3 tools YOUR agent will need (just names + one line each).	45 min	☐
Thu 02 Jul	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 03 Jul	LIVE 5 PM	Show your work. A run log showing a tool call and its result.		★

 **Bring to Friday:** A run log showing a tool call and its result.

Done when:

☐ A tool call fired

☐ You can explain the loop in your own words

☐ Your tool list drafted

WEEK 3

Module 3 — Tools & integrations

Complete before your Fri 10 Jul session · ~3.5 hrs across the week



This week's goal: Your agent uses one REAL tool of yours.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 04 Jul	Watch	Watch Module 3. Note: which of your tools is easiest to add first?	45 min	☐
Sun 05 Jul	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 06 Jul	Build	Add one new tool to <code>agent/tools.py</code> (schema + function). Test it in isolation.	40 min	☐
Tue 07 Jul	Build +	Wire it into the loop; have the agent call it. Office hours if stuck.	40 min	☐
Wed 08 Jul	Make it yours	Point it at YOUR data — your sheet, a CSV export, your files.	45 min	☐
Thu 09 Jul	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 10 Jul	LIVE 5 PM	Show your work. Your agent doing something useful with your own data.		★

 **Bring to Friday:** Your agent doing something useful with your own data.

Done when:

☐ A real tool added

☐ It runs from the agent

☐ It touches your data

WEEK 4

Module 4 — Memory & context

Complete before your Fri 17 Jul session · ~3.5 hrs across the week



This week's goal: Your agent remembers across runs.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 11 Jul	Watch	Watch Module 4. Note: what should your agent remember between runs?	45 min	☐
Sun 12 Jul	Rest / buffer	Catch up if you're behind. Otherwise — recharge.	—	☐
Mon 13 Jul	Build	Use <code>agent/memory.py</code> : have the agent write a note at the end of a run.	40 min	☐
Tue 14 Jul	Build +	On the next run, load that note into the prompt so it 'remembers'.	40 min	☐
Wed 15 Jul	Make it yours	Decide the 1–2 things your agent should remember for your use-case.	45 min	☐
Thu 16 Jul	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 17 Jul	LIVE 5 PM	Show your work. Two runs where run #2 clearly uses what run #1 remembered.		★

📦 Bring to Friday: Two runs where run #2 clearly uses what run #1 remembered.

Done when: ☐ Memory writes ☐ Memory reads back ☐ It's meaningful for your task

WEEK 5

Module 5 — Planning & multi-step

Complete before your Fri 24 Jul session · ~3.5 hrs across the week



This week's goal: Your agent handles a job that takes several steps.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 18 Jul	Watch	Watch Module 5. Note: what is ReAct, in one sentence?	45 min	☐
Sun 19 Jul	Rest / buffer	Catch up if you're behind. Otherwise — recharge.	—	☐
Mon 20 Jul	Build	Give the agent a task needing 3+ tool calls. Watch how it sequences them.	40 min	☐
Tue 21 Jul	Build +	Add a reflection step: let it check its own work before finishing.	40 min	☐
Wed 22 Jul	Make it yours	Write the real 5-step version of YOUR workflow as a single goal.	45 min	☐
Thu 23 Jul	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 24 Jul	LIVE 5 PM	Show your work. A transcript of your agent completing a multi-step task.		★

 **Bring to Friday:** A transcript of your agent completing a multi-step task.

Done when: ☐ 3+ steps in one run ☐ It reached the goal ☐ Your 5-step task written

WEEK 6

Module 6 week — Catch-up & integration (no new video)

Complete before your Fri 31 Jul session · ~3.5 hrs across the week



This week's goal: Your agent does your real job end-to-end, even if rough.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 25 Jul	Watch	No new module. Re-watch any part you were shaky on.	45 min	☐
Sun 26 Jul	Rest / buffer	Catch up if you're behind. Otherwise — recharge.	—	☐
Mon 27 Jul	Build	Connect the pieces: tools + memory + planning in one run.	40 min	☐
Tue 28 Jul	Build +	Run your real workflow's happy path start to finish. Fix what breaks.	40 min	☐
Wed 29 Jul	Make it yours	Tidy: name things clearly, remove dead code, note remaining gaps.	45 min	☐
Thu 30 Jul	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 31 Jul	LIVE 5 PM	Show your work. Your agent doing the real job, start to finish (rough is fine).		★

 **Bring to Friday:** Your agent doing the real job, start to finish (rough is fine).

Done when:

☐ One end-to-end run works

☐ You know your top 2 gaps

☐ Code is readable

WEEK 7

Module 6 — Reliability & guardrails

Complete before your **Fri 07 Aug** session · ~3.5 hrs across the week



This week's goal: Your agent fails safely and never burns money.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 01 Aug	Watch	Watch Module 6. Note: where could your agent do something costly or wrong?	45 min	☐
Sun 02 Aug	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 03 Aug	Build	Add JSON validation + one retry on a tool that can fail.	40 min	☐
Tue 04 Aug	Build +	Set a cost cap in <code>config.py</code> (<code>MAX_USD</code>); add a human confirm step before any 'send'.	40 min	☐
Wed 05 Aug	Make it yours	Add the confirm-before-acting checkpoint that fits YOUR workflow.	45 min	☐
Thu 06 Aug	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 07 Aug	LIVE 5 PM	Show your work. Show it handling a failure gracefully + your cost cap in action.		★

📦 **Bring to Friday:** Show it handling a failure gracefully + your cost cap in action.

Done when:

☐ Validation + retry added

☐ Cost cap set

☐ A human checkpoint exists

WEEK 8

Module 7 — Multi-agent systems

Complete before your **Fri 14 Aug** session · ~3.5 hrs across the week



This week's goal: You know whether your task needs more than one agent.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 08 Aug	Watch	Watch Module 7. Note: when do multiple agents help vs. add chaos?	45 min	☐
Sun 09 Aug	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 10 Aug	Build	Split a job into a planner agent + a worker agent; pass one handoff.	40 min	☐
Tue 11 Aug	Build +	Compare: is the 2-agent version actually better than one? Be honest.	40 min	☐
Wed 12 Aug	Make it yours	Decide if YOUR use-case needs multi-agent — and write why / why not.	45 min	☐
Thu 13 Aug	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 14 Aug	LIVE 5 PM	Show your work. Either a 2-agent handoff working, OR a clear reason one agent is enough.		★

 **Bring to Friday:** Either a 2-agent handoff working, OR a clear reason one agent is enough.

Done when:

☐ You tried a handoff

☐ You made a call for your task

☐ You can justify it

WEEK 9

Module 8 — Deploy your agent

Complete before your **Fri 21 Aug** session · ~3.5 hrs across the week



This week's goal: Your agent runs without you.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 15 Aug	Watch	Watch Module 8. Note: cron vs. GitHub Actions vs. a server — which fits you?	45 min	☐
Sun 16 Aug	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 17 Aug	Build	Add logging so every run records what it did and why.	40 min	☐
Tue 18 Aug	Build +	Schedule it (cron or GitHub Actions). Trigger one scheduled run.	40 min	☐
Wed 19 Aug	Make it yours	Pick the schedule YOUR job needs (daily? hourly? on an event?).	45 min	☐
Thu 20 Aug	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 21 Aug	LIVE 5 PM	Show your work. A log from a run that fired on a schedule, not by you.		★

📦 **Bring to Friday:** A log from a run that fired on a schedule, not by you.

Done when:

☐ Logging works

☐ A scheduled run happened

☐ Schedule matches your need

WEEK 10

Module 9 — Ship a real business agent

Complete before your **Fri 28 Aug** session · ~3.5 hrs across the week



This week's goal: Someone else can run your agent.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 22 Aug	Watch	Watch Module 9. Note: what would confuse a non-technical user?	45 min	☐
Sun 23 Aug	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 24 Aug	Build	Write your agent's playbook: what it does, how to run it, what it needs.	40 min	☐
Tue 25 Aug	Build +	Package the config + a one-line run command. Remove anything fragile.	40 min	☐
Wed 26 Aug	Make it yours	Hand it to ONE non-technical person and watch them try it.	45 min	☐
Thu 27 Aug	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 28 Aug	LIVE 5 PM	Show your work. Feedback from your first real user + a rough 'time saved' number.		★

 **Bring to Friday:** Feedback from your first real user + a rough 'time saved' number.

Done when:

☐ Playbook written

☐ One outside user tried it

☐ You measured time saved

WEEK 11

Capstone build (Week 11)

Complete before your **Fri 04 Sep** session · ~3.5 hrs across the week



This week's goal: Polish your agent for Demo Day.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 29 Aug	Watch	No new module. Re-watch the part most relevant to your capstone.	45 min	☐
Sun 30 Aug	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 31 Aug	Build	Fix your top 3 rough edges. Clean the README so it's obvious how to run.	40 min	☐
Tue 01 Sep	Build +	Do a full dry-run of the real job. Time it.	40 min	☐
Wed 02 Sep	Make it yours	Write your demo story: the problem → your agent → the result.	45 min	☐
Thu 03 Sep	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 04 Sep	LIVE 5 PM	Show your work. A near-final agent + a 2-minute demo plan.		★

📦 **Bring to Friday:** A near-final agent + a 2-minute demo plan.

Done when:

☐ Top 3 issues fixed

☐ README is clean

☐ Demo story written

WEEK 12

Demo Day prep (Week 12)

Complete before your **Fri 11 Sep** session · ~3.5 hrs across the week



This week's goal: A clean, confident 5-minute demo.

DAY	FOCUS	WHAT TO DO	TIME	✓
Sat 05 Sep	Watch	No new module. Watch a peer's <code>#show-your-work</code> post for ideas.	45 min	☐
Sun 06 Sep	Rest / buffer	<i>Catch up if you're behind. Otherwise — recharge.</i>	—	☐
Mon 07 Sep	Build	Rehearse the live run twice. Note where it's slow or risky.	40 min	☐
Tue 08 Sep	Build +	Record a 1-min backup video in case the live run hiccups.	40 min	☐
Wed 09 Sep	Make it yours	Prepare your before/after: how much time your agent saves you.	45 min	☐
Thu 10 Sep	Test & show	Run it, fix one bug, post a screenshot in <code>#show-your-work</code> , and write your #1 question for Friday.	30 min	☐
Fri 11 Sep	LIVE 5 PM	Show your work. Your FINAL working agent — ready to demo on Demo Day. 🎉		★

📦 **Bring to Friday:** Your FINAL working agent — ready to demo on Demo Day. 🎉

Done when:

☐ Rehearsed twice

☐ Backup recording ready

☐ Before/after prepared